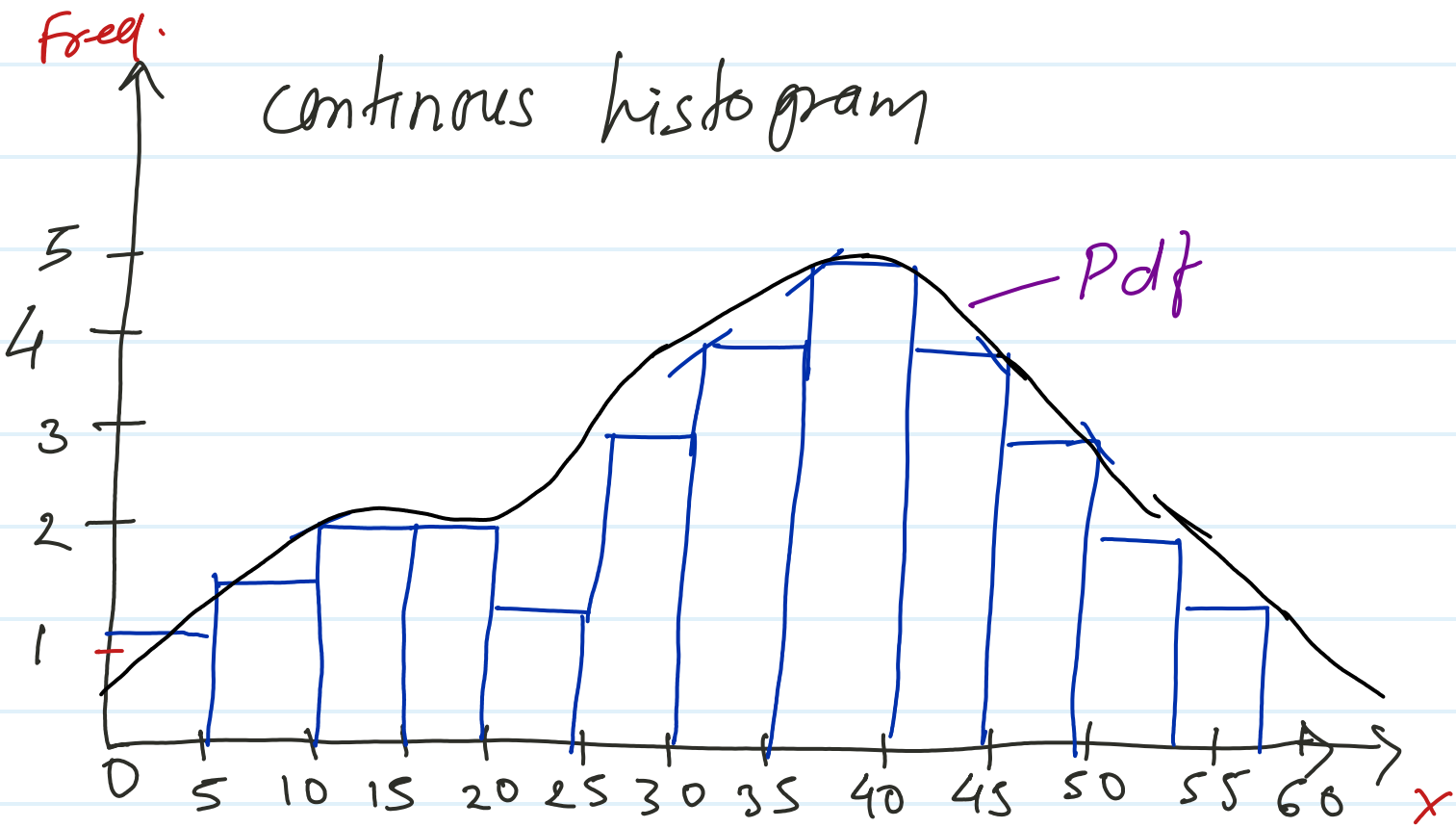
Histogram

[2, 3, 7, 8, 11, 13, 16, 20, 21, 22, 23, 26, 27, 28, 30, 31, 33, 34, 35, 37, 38, 40, 41, 44, 47, 49, 51, 52, 56]

Range = 56 - 2 = 54

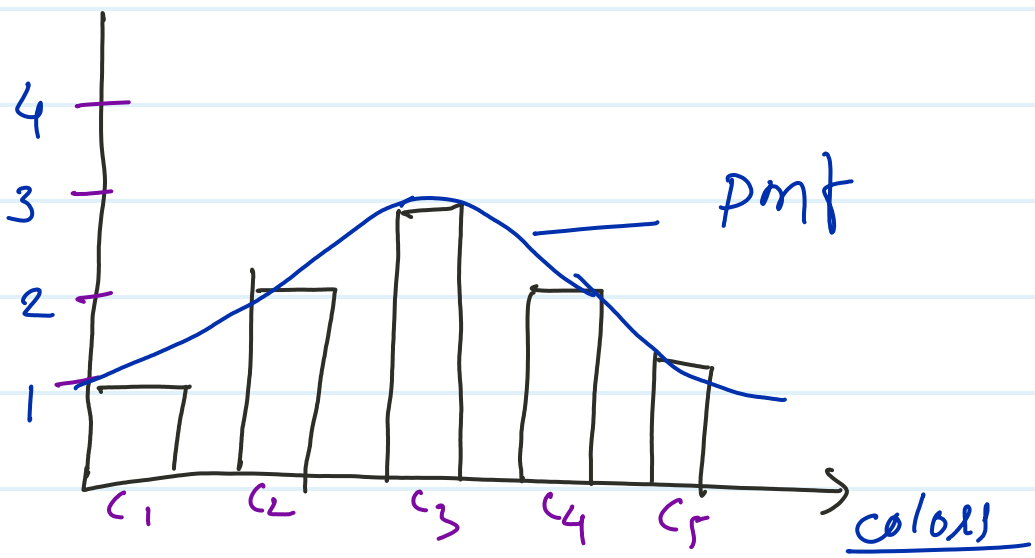
Assume no of bin - 10

Bin size = $\frac{\text{Range}}{\text{no bin}} = \frac{54}{10} = 5.4$



Pdf = Probability density function

Descrete histogram

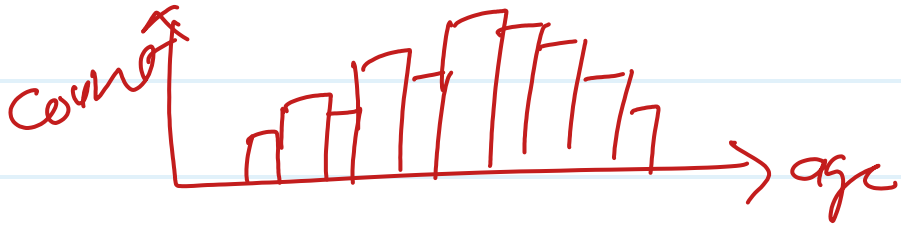


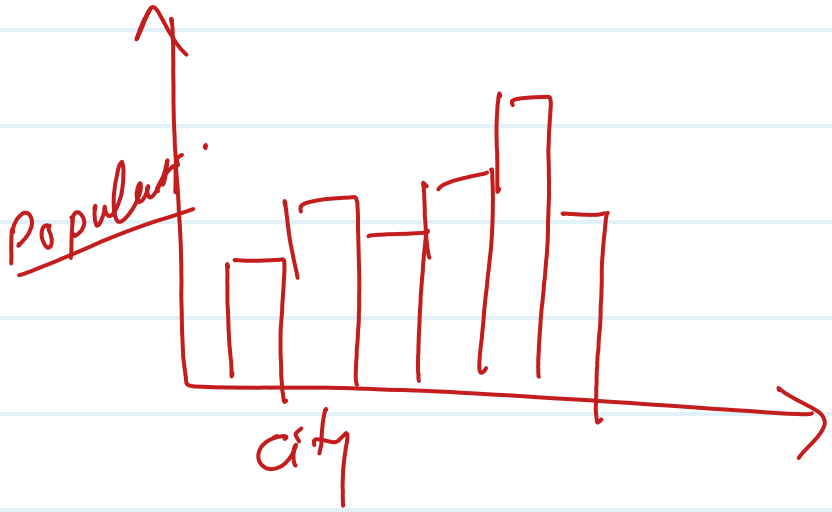
Pmf = Probability mass function.

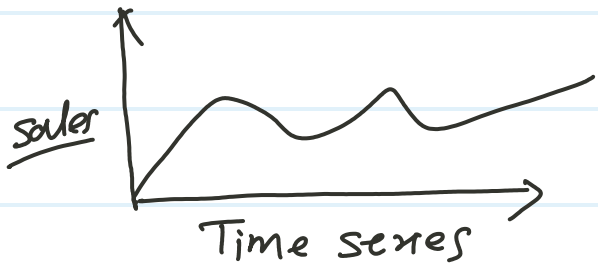
KDE = kernel Density estimator.

X_1	X_2	Y — Target
IQ	CGPA	placement
9	0.75	Y
8	0.45	N
6	0.55	N
4	0.71	Y
3	0.69	Y
5	0.61	Y
6		

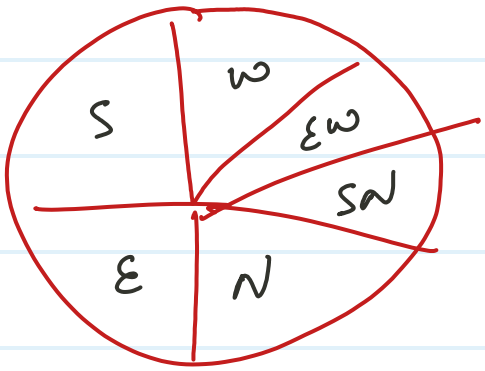
(I) Histogram or Distribution plot
(II) Bar or Column chart
(III) Line chart or Area chart
(IV) Box and whisker plot
(V) pie chart or donut chart
(VI) Heatmap (corr. metric)
(VII) Scatter plot
(VIII) pair plot

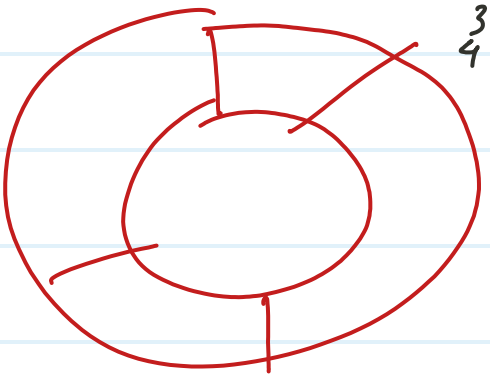


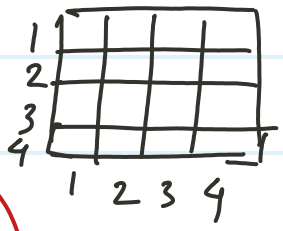




Country sales







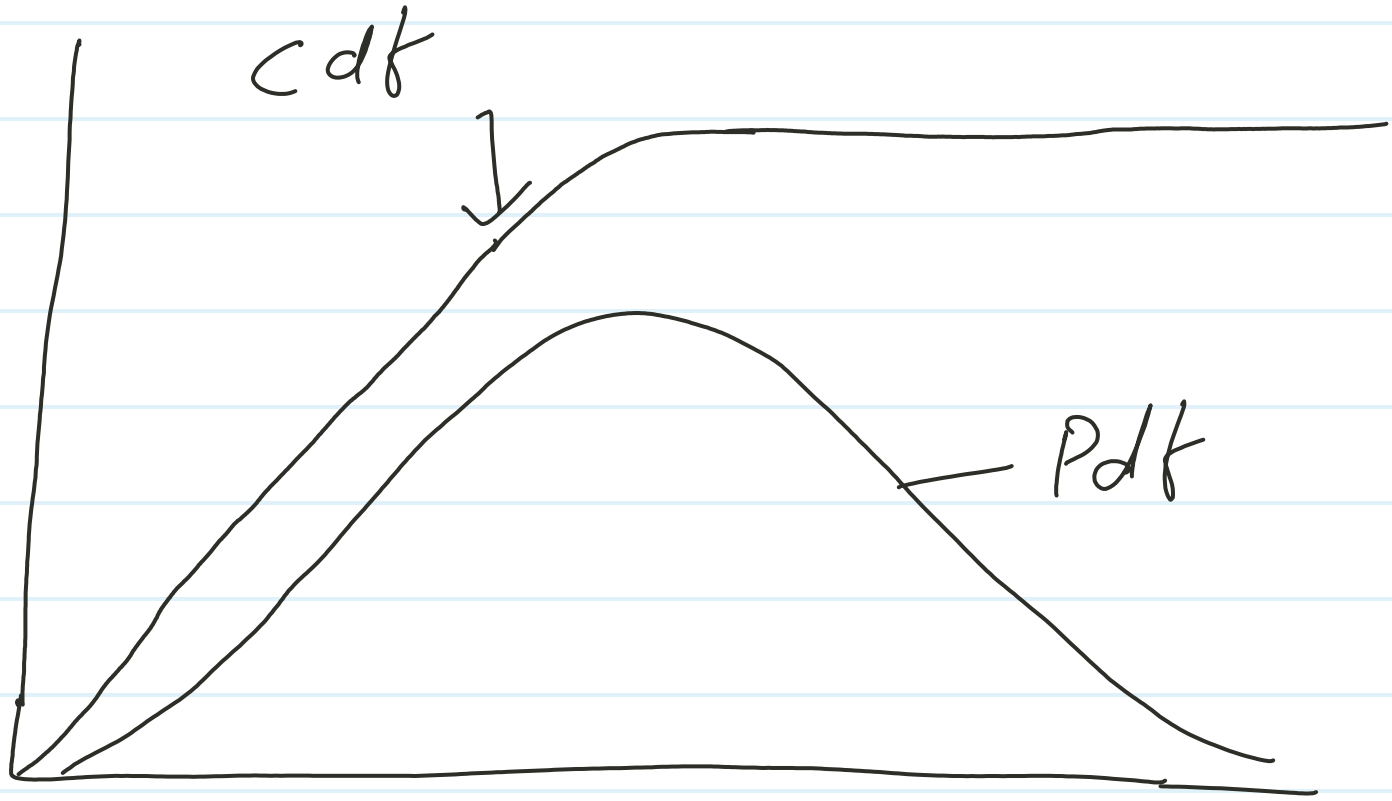
④ column plot

⑤ Pie chart

⑥ Scatter plot

x	y		y'	
1	5		50	
2	10		45	
3	15		40	
4	15	x	40	x
5	20		35	
6	25		30	
7	25	x	25	
8	30		25	x
9	35		20	
10	40		20	x

X		Cummulative
2	3	3
3	2	2+3 = 5
4	1	5+1 = 6
6	2	6+2 = 8
8	4	8+4 = 12
9	3	12+3 = 15
	2	15+2 = 17
	0	17+0 = 17

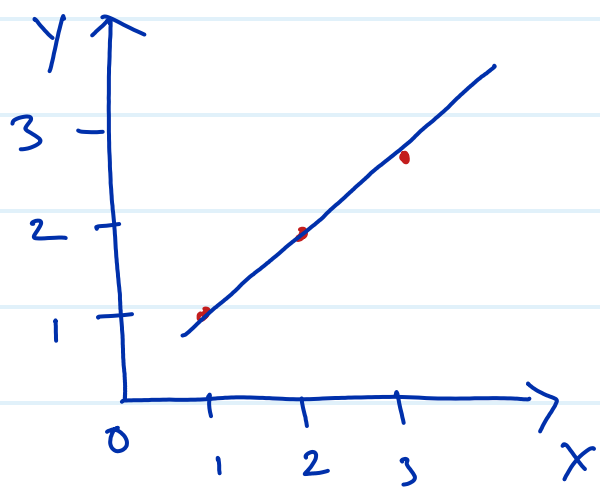


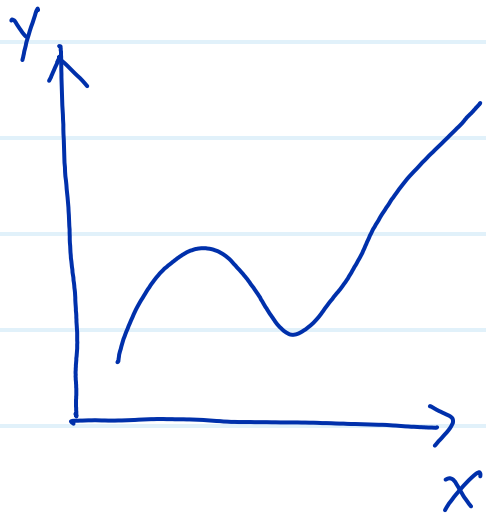
cdf = cummulative Density function.

pop. $\sigma = \sqrt{\dfrac{(x_i - \mu)^2}{N}}$

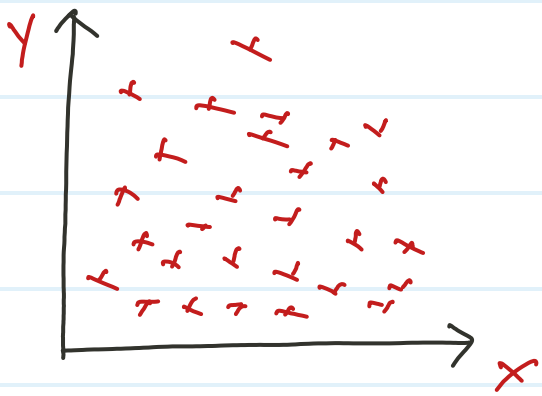
sam $s = \sqrt{\dfrac{(x_i - \bar{x})^2}{n-1}}$

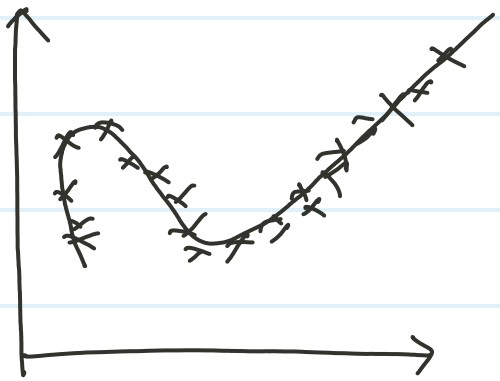
Degree of freedom $= n - 1$





<u>chart</u> → scatter plot





☆ Covariance and correlation.

☆ covariance

X	Y
age	height
2	2
4	2.5
6	3
10	3.5
15	4.5
20	5.5

X	Y
sharemarket	GDP
10k	2.3
20k	3.1
30k	3.9
40k	4.8
50k	5.2

$X \uparrow \quad Y \uparrow \rightarrow$ Positive

$X \downarrow \quad Y \downarrow \rightarrow$ Positive

$X \uparrow \quad Y \downarrow \rightarrow$ Neg

$X \downarrow \quad Y \uparrow \rightarrow$ Neg.

x_1

$X_1 \uparrow \quad Y$

$X_1 \downarrow \quad Y$

$X_1 \uparrow \quad Y \uparrow$

$X_1 \uparrow \quad Y \downarrow$

zero covariance

Age	weight	height	BMI
1	2	3	4

Independent

Target

$1 = 4$
$2 = 4$
$3 = 4$

+ / -

$1 = 2$
$2 = 3$
$1 = 3$

zero covariance

$$Cov(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

limit $(+\infty, -\infty)$

Correlation

① pearson's correlation coefficient
② spearman Rank corr coeff.

① Pearson's correlation coefficient

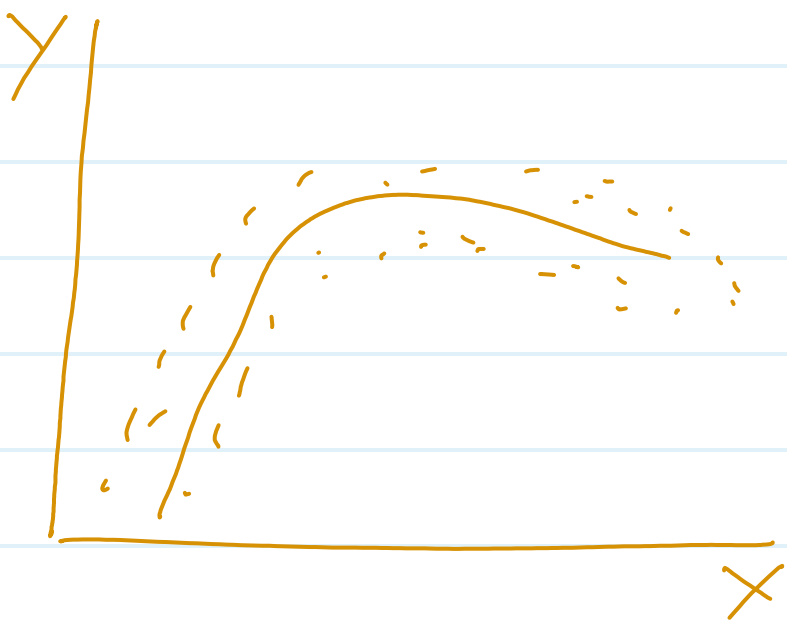
$$Cor(X, Y) = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y}$$

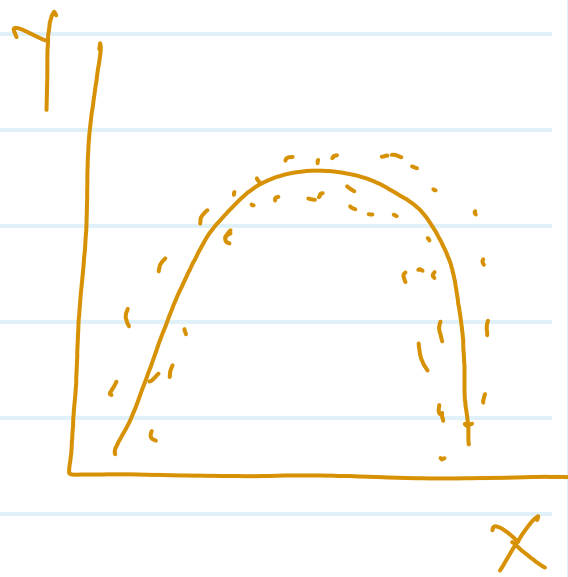
limit $(-1, +1)$

Disavantage -

Can't identified non-linear relation

+/-





(II) Spearman Rank corr coff.

$$Cor_R(X, Y) = \frac{6\sum d_i^2}{n(n^2-1)}$$

Eg

X	R_X	Y	R_Y	$d_i = R_X - R_Y$
4	3	10	6	$= -3$
5	2	13	3	$= -1$
6	1	11	5	$= -4$
3	4	15	2	$= 2$
1	6	16	1	$= 5$
2	5	12	4	$= 1$

$$d = (-3)^2 + (-1)^2 + (-4)^2 + (2)^2 + (5)^2 + (1)^2$$
$$= 9 + 1 + 16 + 4 + 25 + 1 \Rightarrow 56$$

$$\frac{6\sum d_i^2}{n(n^2-1)} = \frac{6 \times 56}{6(6^2-1)} = \frac{6 \times 56}{6 \times 35}$$

$\Rightarrow$

Covariance

$$\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

X	Y
5	50
10	60
20	75
15	70
25	85

$$\bar{x} = \frac{75}{5} = 15$$

$$\bar{y} = \frac{340}{5} = 68$$

$x_i - \bar{x}$	$y_i - \bar{y}$
$5 - 15 = -10$	$50 - 68 = -18$
$10 - 15 = -5$	$60 - 68 = -8$
$20 - 15 = 5$	$75 - 68 = 7$
$15 - 15 = 0$	$70 - 68 = 2$
$25 - 15 = 10$	$85 - 68 = 17$

$$\frac{\sum (-10 + (-5) + 5 + 0 + 10) \cdot (-18 + (-8) + 7 + 2 + 17)}{5-1}$$

$$= \frac{30 \times 52}{4} \Rightarrow \underline{\underline{390}}$$

X	Y
10 ↑	?

$$X, Y = 390$$

$$\sigma_x = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

$$= \sqrt{\frac{1}{5-1} (-10)^2 + (-5)^2 + 5^2 + 0 + (10)^2}$$

$$\Rightarrow \sqrt{\frac{100 + 25 + 25 + 100}{4}}$$

$$\Rightarrow \sqrt{\frac{250}{4}} = \sqrt{62.5}$$

$$\sigma_x = 7.90$$

$$\sigma_y = \sqrt{\frac{(-18)^2 + (-8)^2 + 7^2 + 2^2 + 17^2}{4}}$$

$$= \sqrt{\frac{324 + 64 + 49 + 289}{4}} \Rightarrow$$

$$= \sqrt{\frac{730}{4}} = \sqrt{182}.$$

$$\sigma_Y \Rightarrow 13.5$$

$$\Rightarrow \frac{390}{7.90 \times 13.5} \qquad \frac{390}{106}$$

$$\Rightarrow \underline{\underline{0.365}}$$

$$-1 \quad — \quad +1$$

$$\mathrm{cov}(X,Y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$SD_x = \frac{1}{n-1} \sqrt{\sum (x_i - \bar{x})^2}$$

$$SD_y = \frac{1}{n-1} \sqrt{\sum (y_i - \bar{y})^2}$$

$$= \frac{\cancel{\frac{1}{n-1}} (x_i - \bar{x})(y_i - \bar{y})}{\cancel{\frac{1}{n-1}} \left(\sqrt{(x_i - \bar{x})^2} \times \sqrt{(y_i - \bar{y})^2} \right.}$$

$$\Rightarrow \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(x_i - \bar{x})^2}\ \sqrt{(y_i - \bar{y})^2}}$$

* Spearman correlation Coefficient

$$\rho = \frac{6 \sum d_i^2}{n(n^2-1)}$$

X		Y		diff	$(d_i)^2$
20	1	10	1	0	0
30	3	15	2	1	1
40	4	25	4	0	0
25	2	20	3	-1	1
50	5	35	5	0	0
					2

$$= \frac{6 \times 2}{5(5^2-1)} = \frac{\cancel{12}\ 1}{\cancel{120}\ 10}$$

$$= \underline{\underline{0.01}}$$